

Outline

- **Cyber-Physical Systems**
 - **Cyber-Physical WAMPAC**
 - **Cyber-Physical Water Treatment System**

Cyber Physical Systems (CPS)

Intra-Car

**Engine Control:
Electronically
controlled**
- Injection
- Compression
- Combustion
- ...

**Automatic
Gearbox**

Climate Control

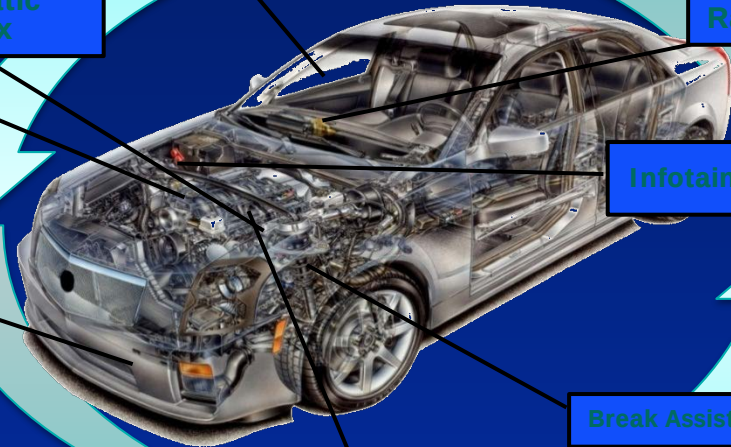
Rain Sensors

Infotainment System

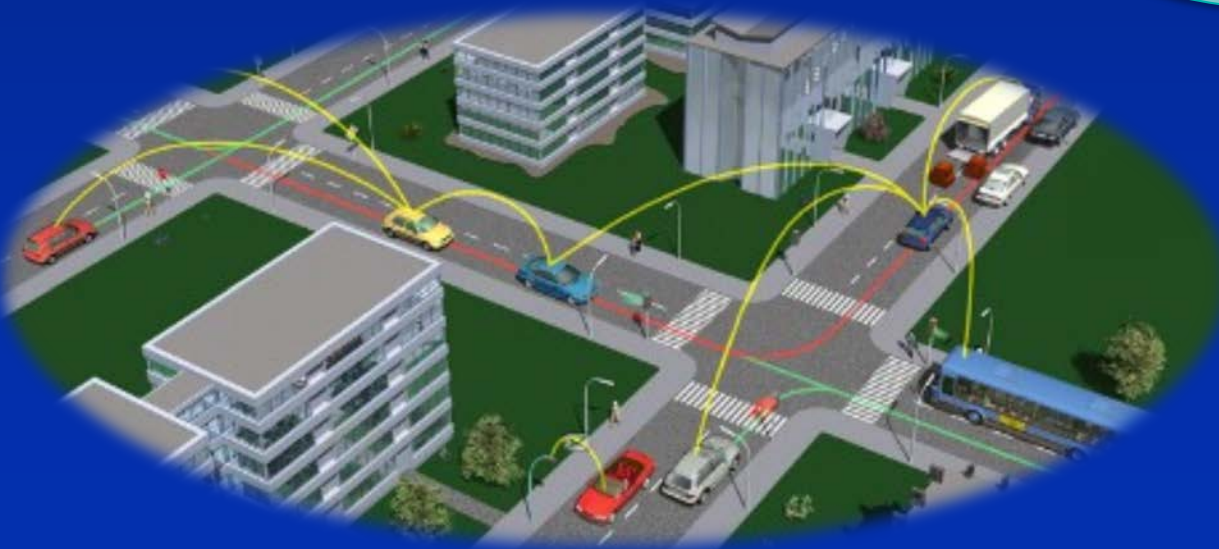
**Park Distance
Control System**

Break Assist System

**Electrical Drive
Engineering, e.g.
Hybrid Drive**



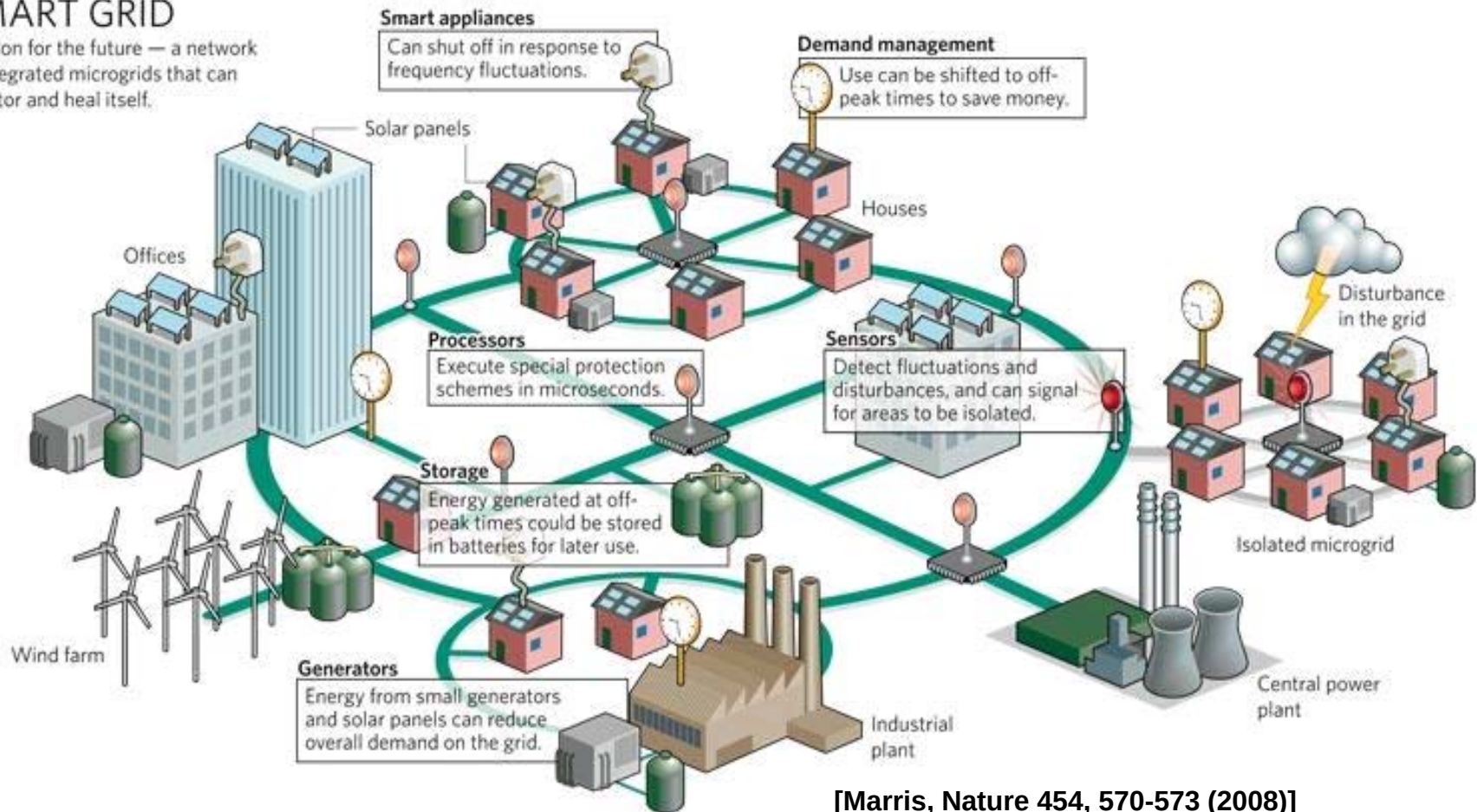
Inter-Car



Cyber Physical Systems (CPS)

SMART GRID

A vision for the future — a network of integrated microgrids that can monitor and heal itself.



[Marris, Nature 454, 570-573 (2008)]

Outline

- CPS concept (definitions, characteristics, properties)
- **CPSs application domains**
- **CPSs research challenges**
 - **Design principles**
 - **Modeling**
 - **Generic architecture**
- **Cyber-Physical WAMPAC**
- **Cyber-Physical Water Treatment System**

CPS Concept

- **Technological advances from the recent decades => early examples of a new systems generation**
 - **Quality, safety and efficiency critical infrastructure systems**
 - **Integrated, self-optimizing transportation systems**
 - **Air planes and automobiles that are environmentally friendly and energy efficient**
 - **Advanced healthcare via increased automation, integrating smart devices, and providing safe access to electronic medical records**

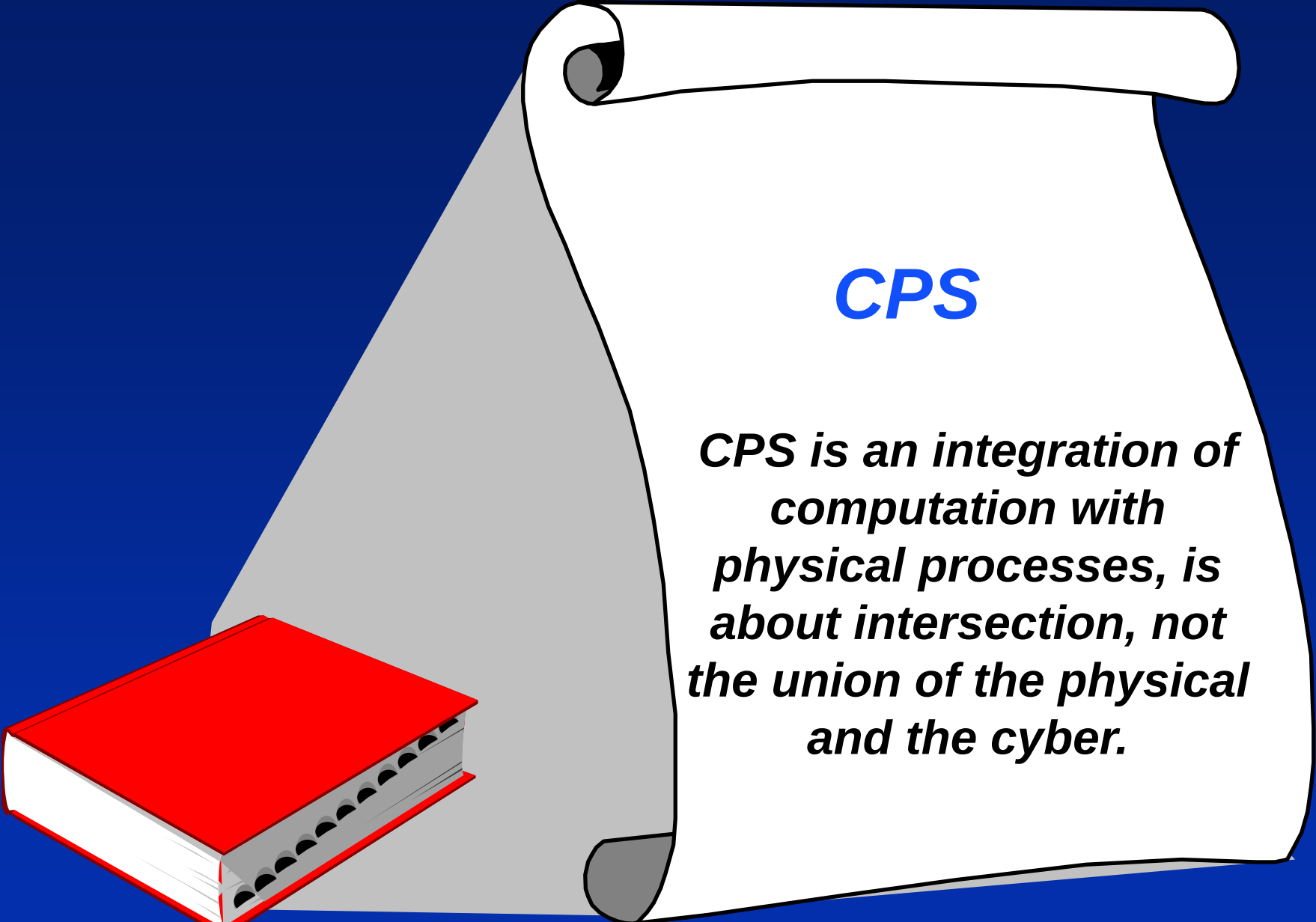
CPS Concept

- **CPSs = a new generation of systems representing more than networking and information technology, information and knowledge being integrated into physical objects**
- **CPSs = a new generation of intelligent and autonomous systems (CPS-Steering-Group, 2008))**

CPS

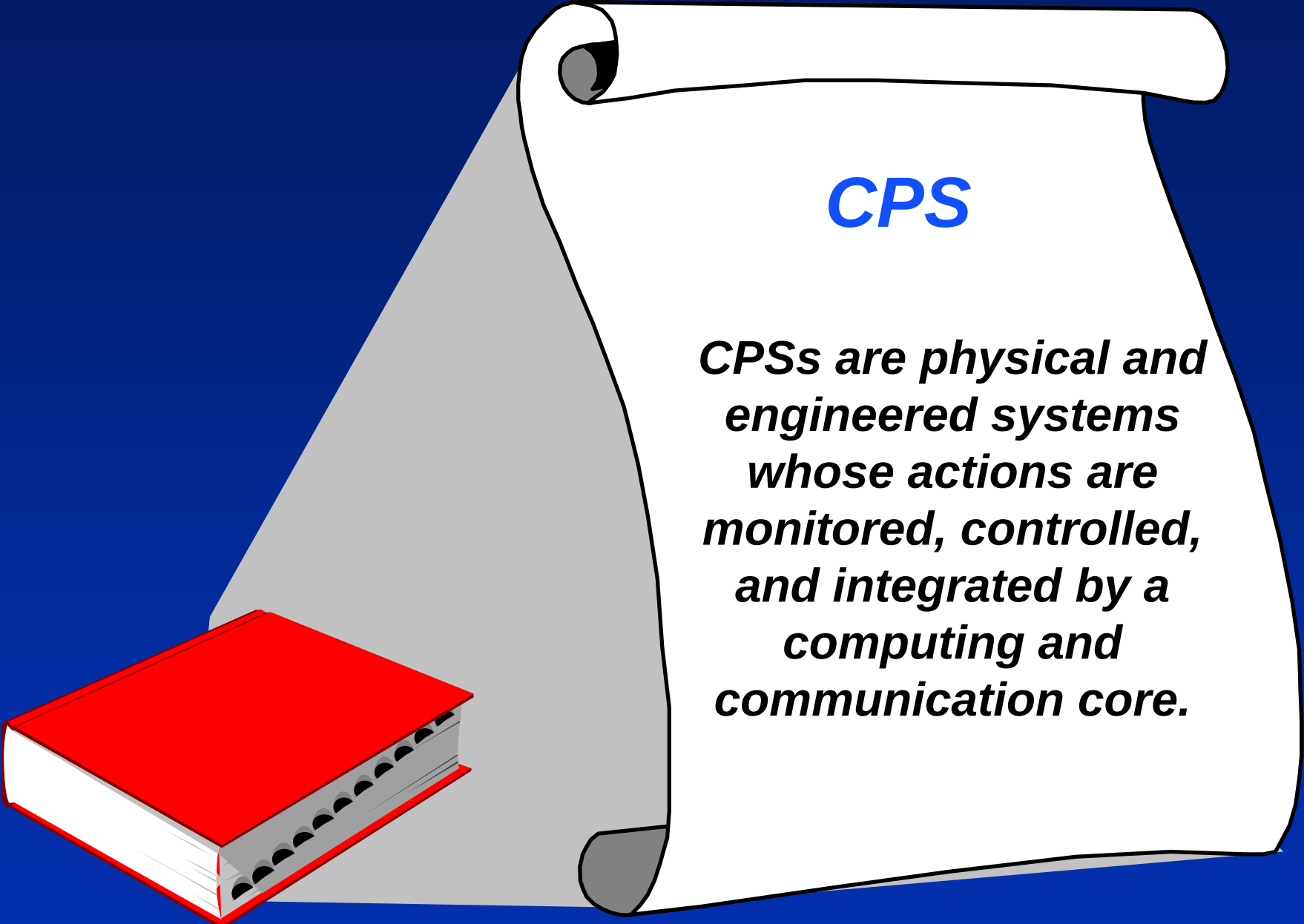
A CPS integrates computing, communication and storage capabilities with monitoring and/or control of entities in the physical world, and must do so dependably, safely, securely, efficiently and real-time.

***S. Sastry,
University of California,
Berkeley***



CPS

CPS is an integration of computation with physical processes, is about intersection, not the union of the physical and the cyber.



CPS

CPSs are physical and engineered systems whose actions are monitored, controlled, and integrated by a computing and communication core.

CPS Concept

- **CPSs are not**
 - Traditional embedded systems or the actual real-time systems
 - Today's sensor networks
 - Desktop applications
- **CPSs present**
 - Cyber capabilities in every physical component
 - Dynamically reconfiguring/reorganizing
 - High degrees of automation
 - Operation must be dependable and certified in some cases
 - Cyber and physical components are integrated for learning and adaptation, higher performance, self-organisation, auto-assembly

CPS Concept

- **CPSs main features**
 - **Input and feedback from/to the physical environment - the existence of the secured communication channels**
 - **Management and distributed control - a federated approach**
 - **Real-time performance requirements**
 - **Large geographical distribution**
 - **Very large scale control systems (SoS)**

CPS Concept

- **CPSs properties**

- **Functionality**
- **Performance**
- **Dependability**
- **Security**
-
- **Cost**

Usability

Administration

Adaptability

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CPSs Application Domains

- Critical infrastructure control
- Safe and efficient transport
- Alternative energy
- Environmental control
- Medical devices
- Telemedicine
- Assisted living
- Social networking
- Manufacturing
- Agriculture



CPSs Application Domains

CPSs for the automotive industry requires high computing power, due to complex traffic control algorithms that calculate for example the best route according to traffic situation.



K. Wan, K.L. Man, and D. Hughes, Specification, Analyzing Challenges and Approaches for Cyber-Physical Systems (CPS), Engineering Letters, 18(3), 2010

CPSs Application Domains

CPSs for environment monitoring, distributed in a wide and varied geographical area must operate without human intervention for long time periods with minimal energy consumption.

In such an environment, the accurate and in time data collection provided by the ad-hoc network with low power consumption, represent a real research challenge.



K. Wan, K.L. Man, and D. Hughes, Specification, Analyzing Challenges and Approaches for Cyber-Physical Systems (CPS), Engineering Letters, 18(3), 2010

CPSs Application Domains

CPSs for aviation and defense require a precise control and high security and not least high power computing.

In this scope, the development of the security protocols will be the main research challenge.



K. Wan, K.L. Man, and D. Hughes, Specification, Analyzing Challenges and Approaches for Cyber-Physical Systems (CPS), Engineering Letters, 18(3), 2010

CPSs Application Domains

CPSs for energy control, water resources management, etc. require a precise and reliable control, leading to application software methodologies to ensure the quality of the software.



K. Wan, K.L. Man, and D. Hughes, Specification, Analyzing Challenges and Approaches for Cyber-Physical Systems (CPS), Engineering Letters, 18(3), 2010

CPSs Application Domains

CPSs for healthcare and medical equipments require a new generation of analysis, synthesis and integration technologies, leading to the development and the application of the interoperability algorithms.



K. Wan, K.L. Man, and D. Hughes, Specification, Analyzing Challenges and Approaches for Cyber-Physical Systems (CPS), Engineering Letters, 18(3), 2010

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CPSs Research Challenges

- ***Abstraction and Architectures*** - to enable control, communication and computing integration for the rapid design and implementation of CPSs
- ***Distributed Computations and Networked Control*** – new algorithms, methods and tools to meet the high reliability and security requirements of cooperative heterogeneous components that interact with complex physical environment
- ***Verification and Validation*** - hardware and software components as well as the systems they form, have to achieved a high degree of dependability

CPSs Research Challenges

	Aviation Defense	Auto	Energy	Healthcare
New abstraction layers for design	BI	BI	BI	BI
Semantic foundations for composing models	BI	BI	I	I
Composition platforms for heterogeneous systems	BI	BI	BI	BI
Foundation for system integration	BI	BI	BI	BI
Infrastructure for automatic design	BI	BI	LI	I
Open architectures	I	I	BI	I
Dependability and security	BI	BI	BI	BI

BI-Big Importance, I-Importance, LI-Low Importance

CPS-Steering-Group, Cyber-physical systems - executive summary,
<http://varma.ece.cmu.edu/summit>, 2008

CPSs Research Challenges - Design

- **Design decomposition by orthogonal aspects of the problem**
- **Strong interdependences between the CPSs components limit the compositionality in the design process**
- **Abstraction of the system interdependencies is needed and this procedure can be achieved by the development of new modeling techniques => *Model-based design (Modellica)***

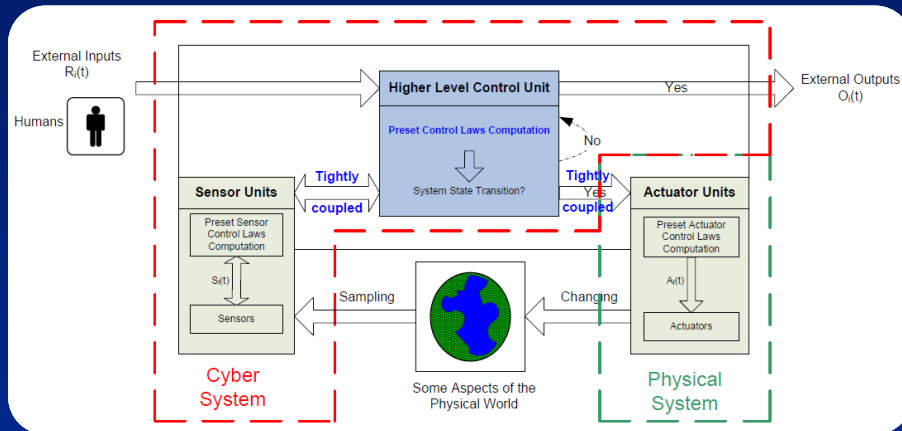
J. Sztipanovits, Composition of Cyber-Physical Systems, Proceedings of the 14th Annual IEEE International Conference and Workshops on the Engineering of Computer-Based Systems ECBS'07, 2007

CPSs Research Challenges - Modeling

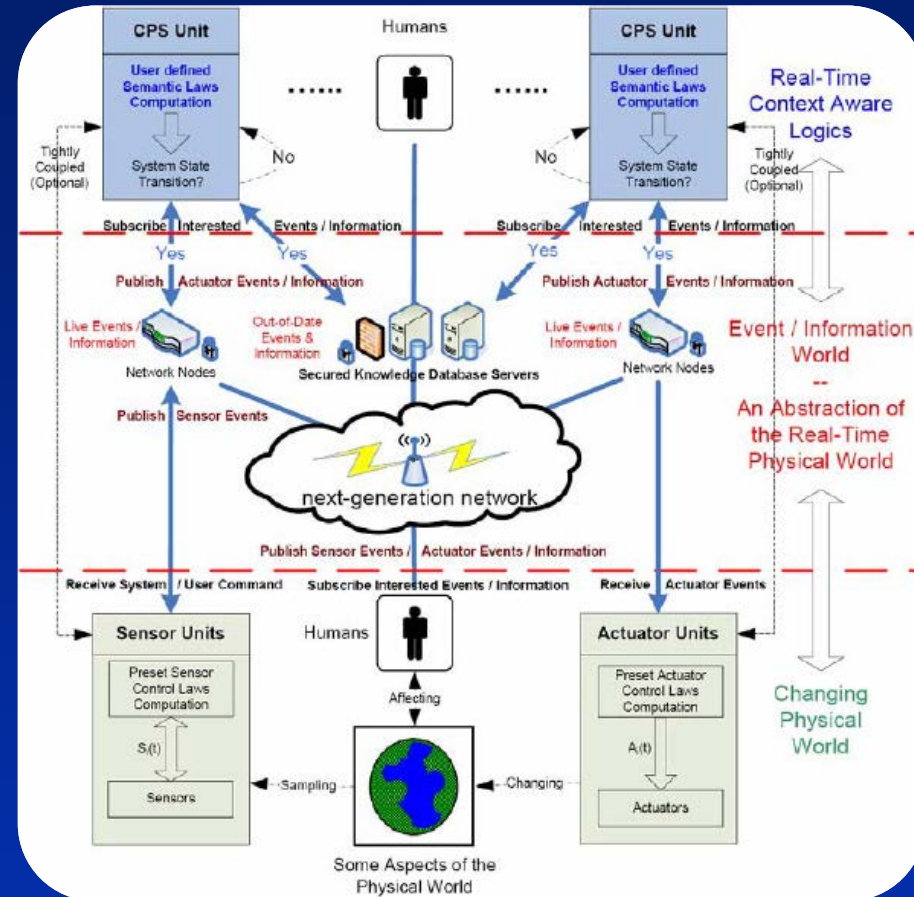
- **Meta-modeling**
- **Semantic modeling - as foundation for CPSs semantics**
 - **Event-based semantics**
 - **Ontology-based semantics**
 - **Two compositional models implemented with multi-agent techniques**
 - **One for autonomous agents**
 - **Other for interactive agents - seems to provide the forms of interaction such as needed in CPSs**

J. Sztipanovits, Composition of Cyber-Physical Systems, Proceedings of the 14th Annual IEEE International Conference and Workshops on the Engineering of Computer-Based Systems ECBS'07, 2007

CPSs Research Challenges - Architecture



- **Embedded systems architecture**
- **CPS prototype architecture**
 - **Global Reference Time**
 - **Event/Information Driven**
 - **Quantied Condence**
 - **Publish/Subscribe Scheme**
 - **Semantic Control Laws**
 - **New Networking Techniques**



Y. Tan, S. Goddard, and L.C. Perez,
A prototype Architecture for Cyber-Physical Systems, SIGBED Review,
5(1), 2008

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Cyber-Physical Wide-Area Monitoring, Protection & Control (Cyber-Physical WAMPAC)

- Electric power systems are the largest machines ever built
- Systems are vast in terms of:
 - Geographic distance
 - Amounts of MW transported
 - Number of domains they span
 - Time-domain (μ s TO days) – equal importance for stability and reliability
 - IT and communication (real-time embedded controllers TO back-office transactions based enterprise systems)
 - Organizational – several entities (ISOs, TSOs, DSOs, GenCo's)

Cyber-Physical WAMPAC open topics

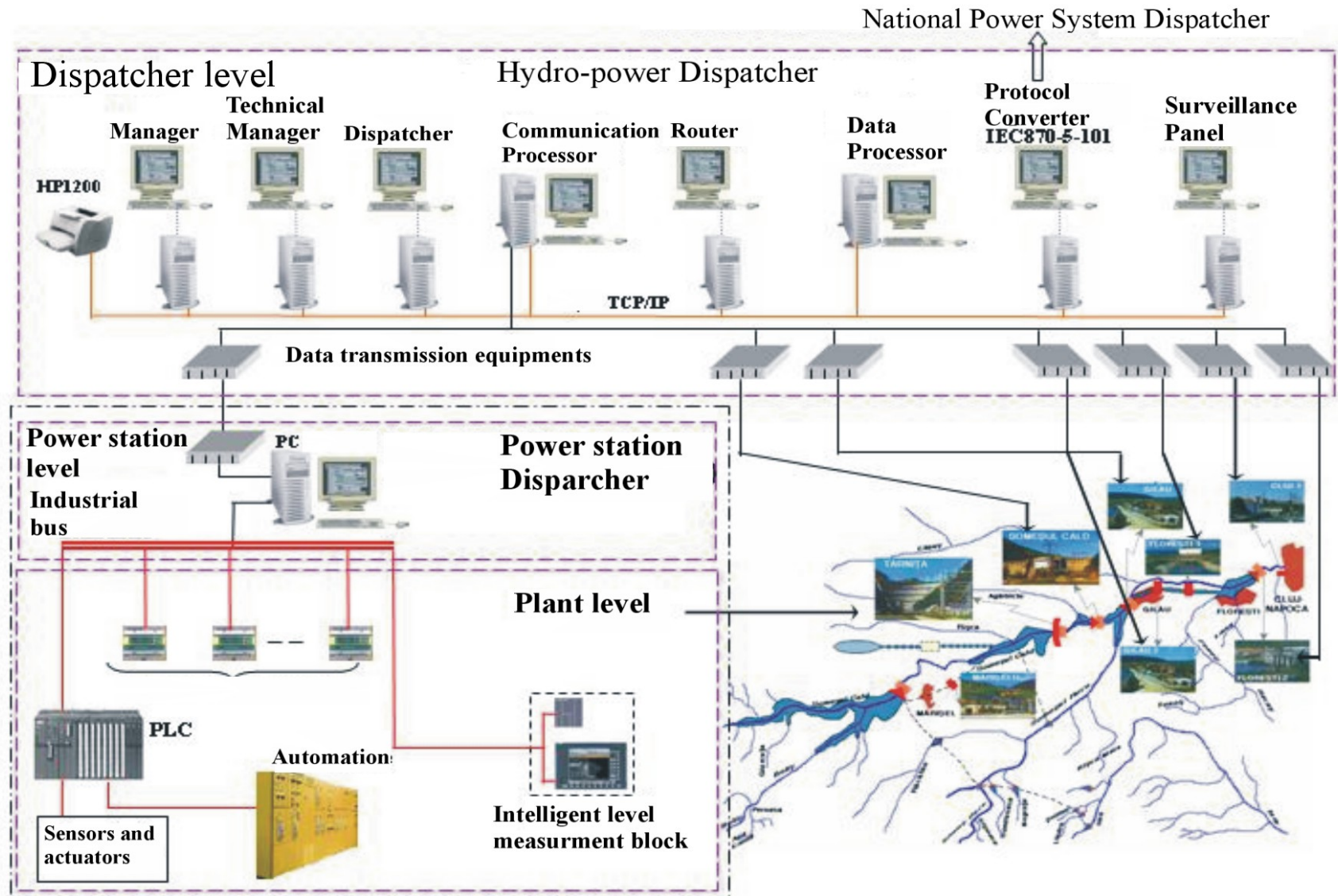
- **WAMPAC architecture and concepts**
- **Architectural aspects of Wide Area data exchange**
 - Quality of Service schemes for data delivery
 - Peer to Peer architectures for real-time data exchange
 - Overlay networks for data exchange
 - Middleware for real-time data access
- **WAMPAC dependencies on Communication**
 - Characterizations of power grid dependencies on ICT
 - Affect on power schemes of performance problems or failures in ICT: message drops, over-budget latencies, complete failures of networking or databases, etc
- **Interoperability of WAMPAC systems**
- **Control center aspects of WAMPAC**
 - Real time visualization of power systems
 - State estimation using information of different control centers
 - Data Mining and analytics



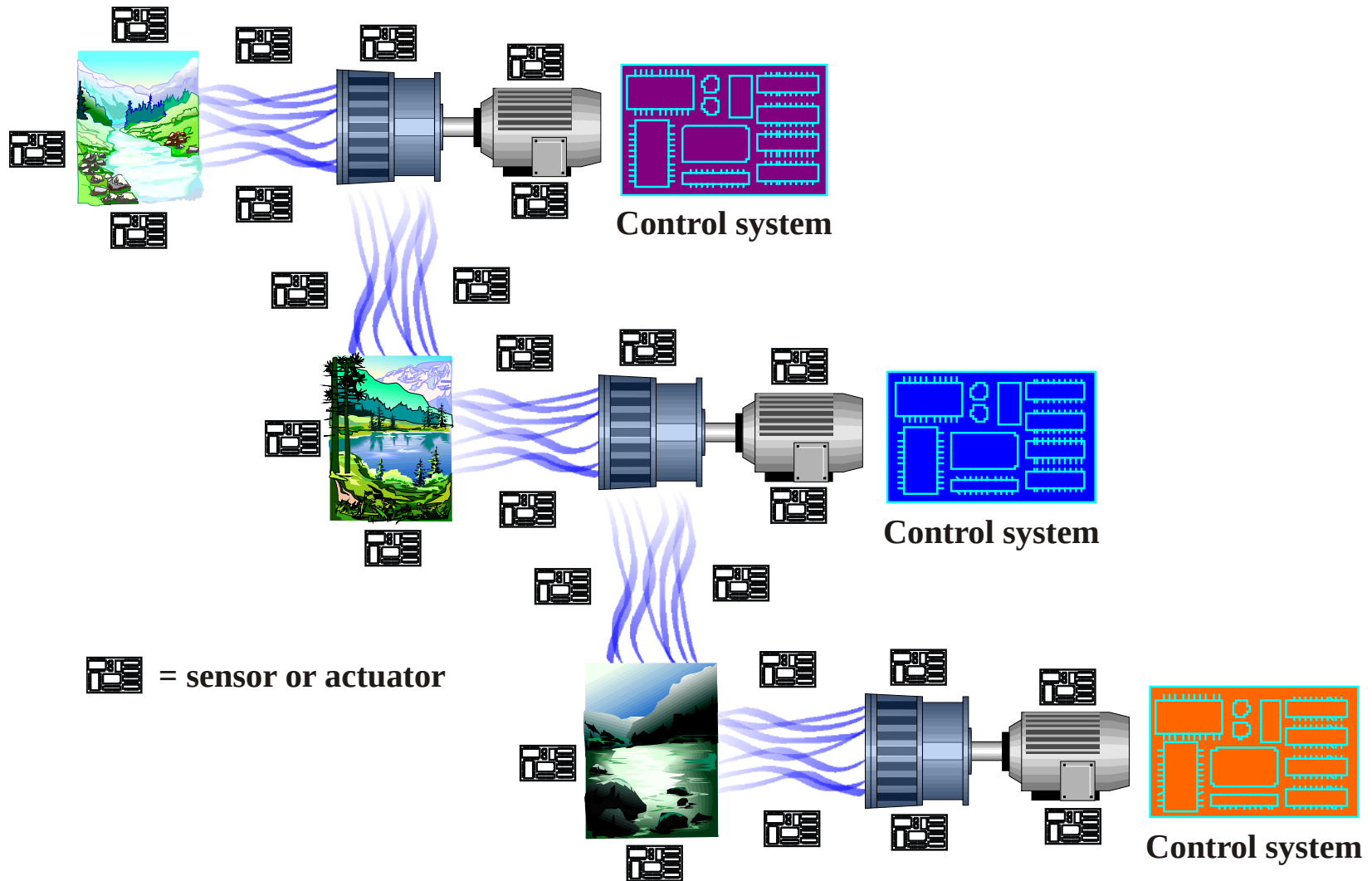
HYDROPOWER PLANTS CHAIN CPS - *is a CPS from hydropower production field that incorporates a number of monitoring and control devices to act as a distributed, fault-tolerant, and real time constrained control system to cooperatively adjust power flows*

Hydropower plants chain CPS characteristics:

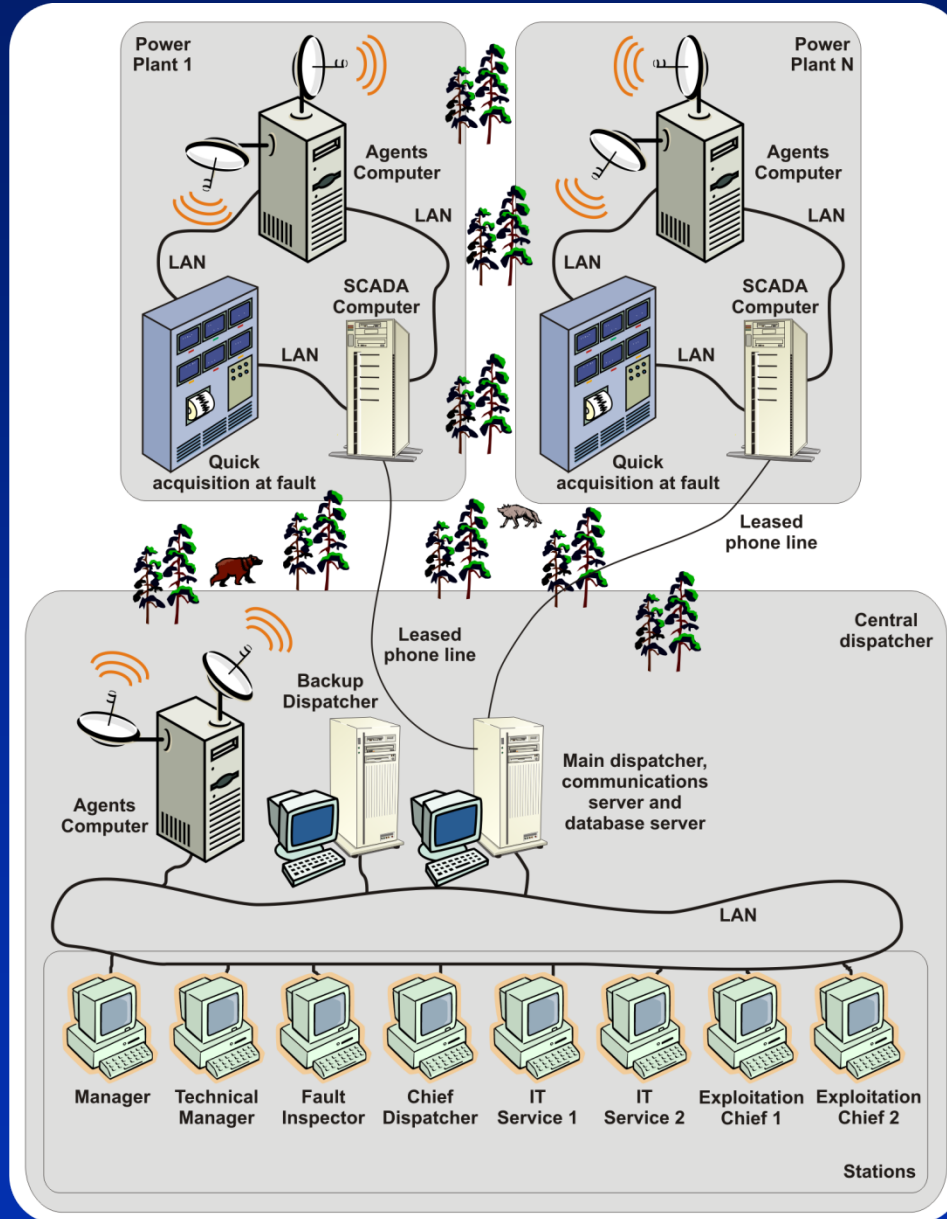
- Controls and monitors the hydropower production process;
- Up and down openness;
- Adaptability;
- Real time operating;
- Distributed on a wide geographical area.



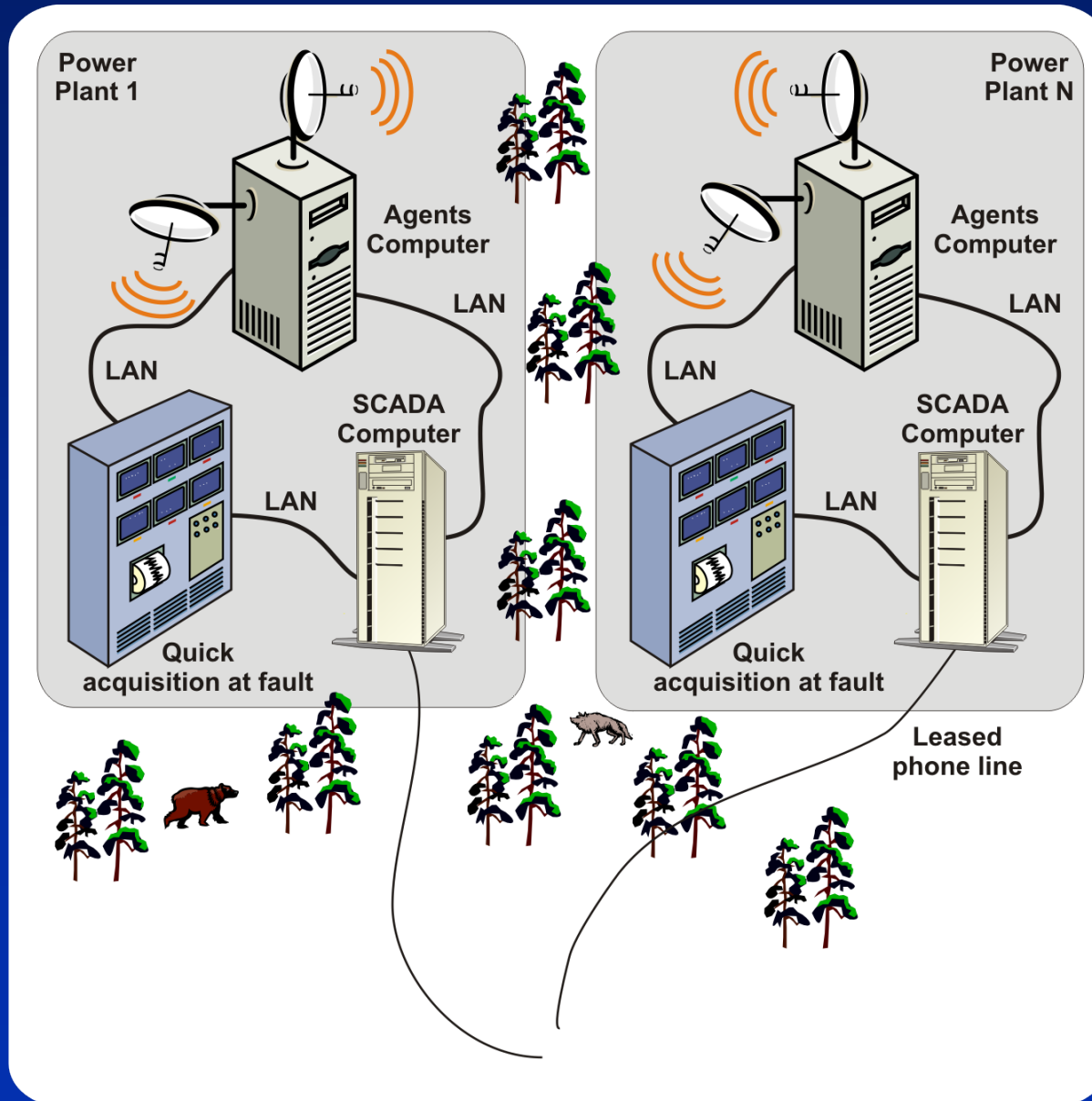
Hydropower plants chain structure



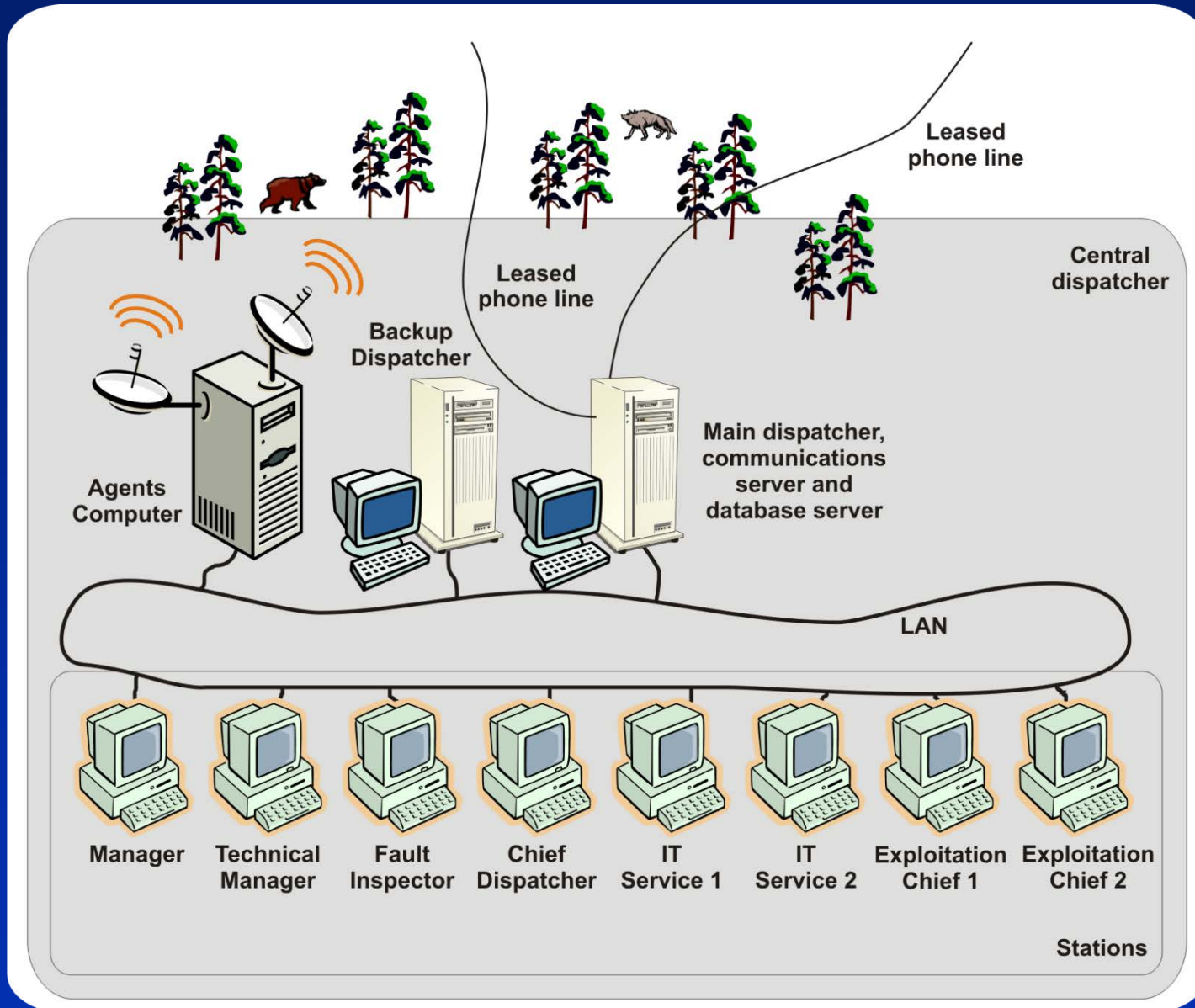
CPS structure



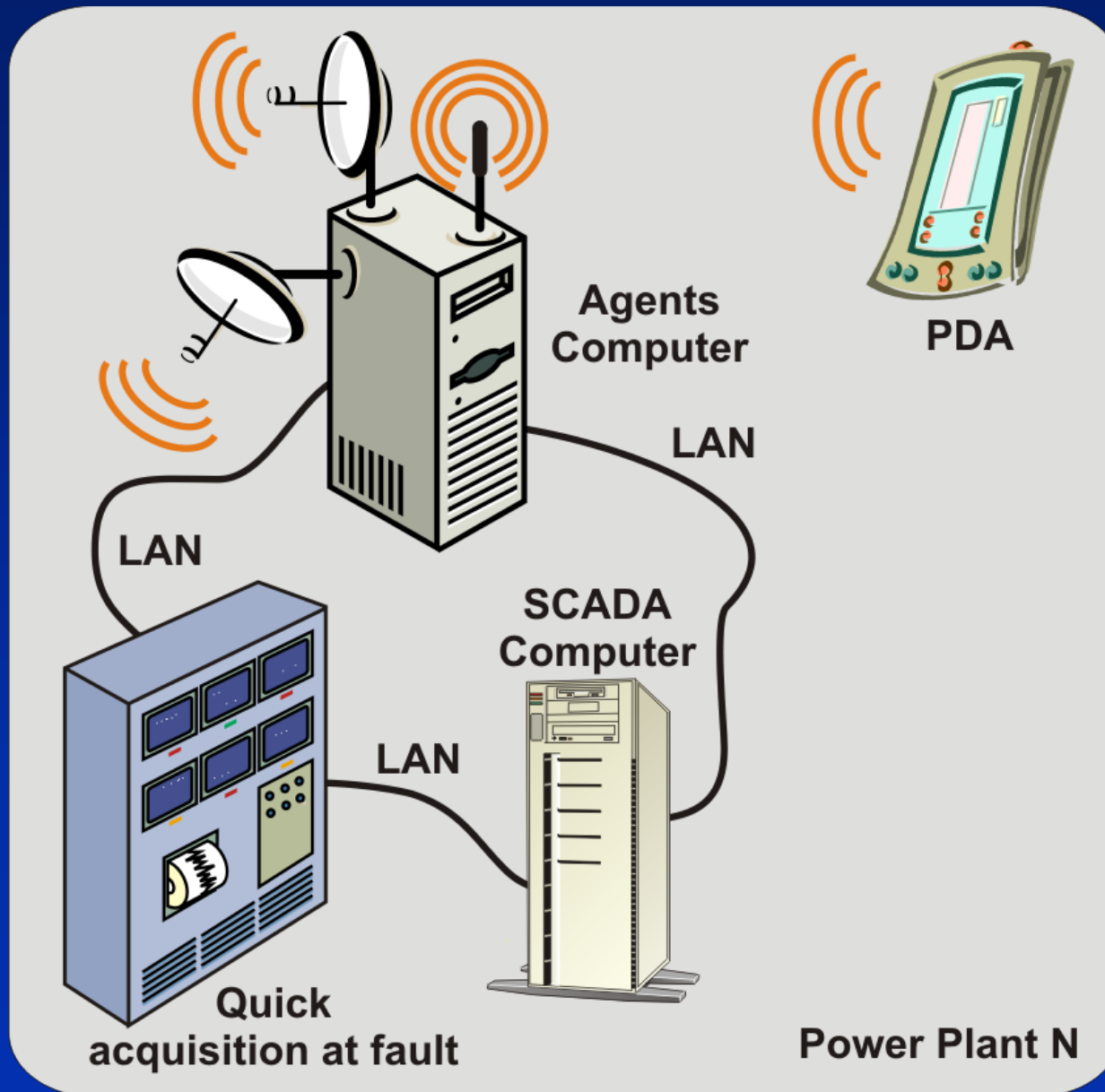
CPS plant site



CPS dispatcher site



PDA access



Partial view of a dispatcher level graphical interface (Java)

Online Tarnita

Grup HG1

P	Q	U	I mediu	I faza R	I faza S	I faza T
0.288 [MW]	0.135 [MVAR]	2.54 [KV]	248.0 [A]	223.0 [A]	150.0 [A]	127.0 [A]

Grup HG2

P	Q	U	I mediu	I faza R	I faza S	I faza T
0.126 [MW]	0.156 [MVAR]	1.47 [KV]	268.0 [A]	274.0 [A]	169.0 [A]	254.0 [A]

TSP 1

U	I
1.56 [KV]	179.0 [A]

TSP 2

U	I
1.68 [KV]	180.0 [A]

TH 1

U	I
1.71 [KV]	182.0 [A]

TH 2

U	I
1.54 [KV]	187.0 [A]

T3-110KV

F	I
147.0 [Hz]	183.0 [A]

T3-20KV

U	I
1.05 [KV]	181.0 [A]

TS3-20KV

U	I
1.32 [KV]	177.0 [A]

LEA 1 Agpires

U	I
1.59 [KV]	184.0 [A]

LEA 2 Floresti

U	I
1.65 [KV]	153.0 [A]

Nivel amonte **Nivel aval**

287.0 [m]	163.0 [m]
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Visualizer-generated configuration pages

PLC-S26-Tarnita

☐ Testable

[Select tests](#)

☐ Log test result to db

☐ Reset if no response

Test freq sec

[Save](#)

for a PLC

OPG1

Test

☐ Voltage

☐ Current

☐ Digital

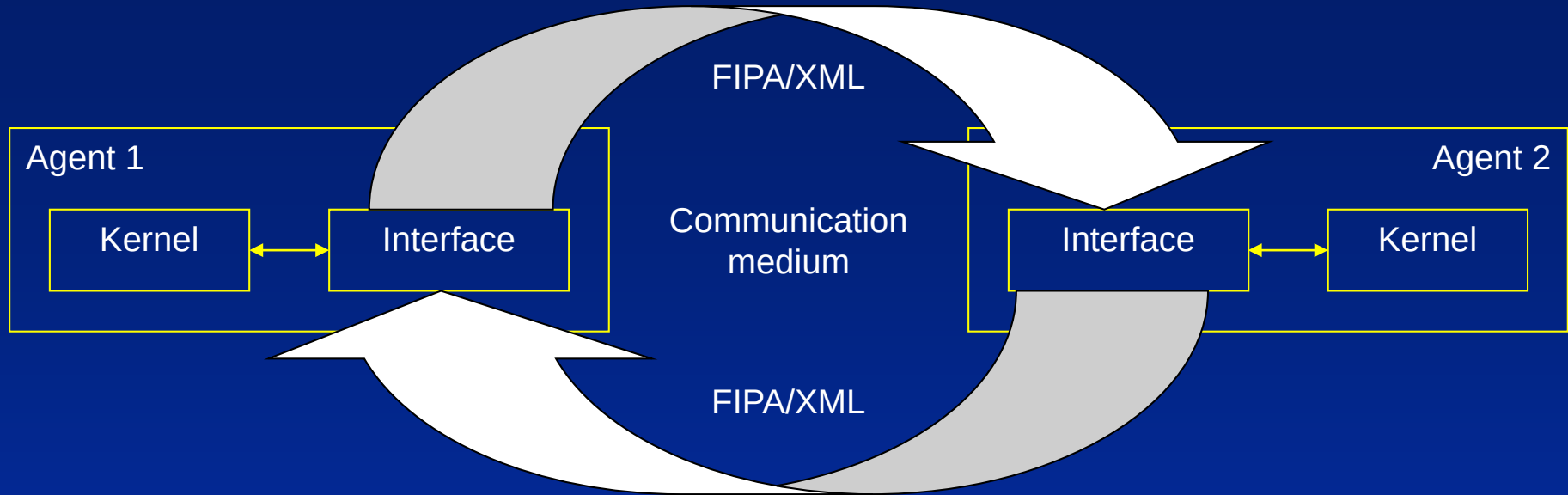
☐ Log test result to db

☐ Reset if no response

[Save](#)

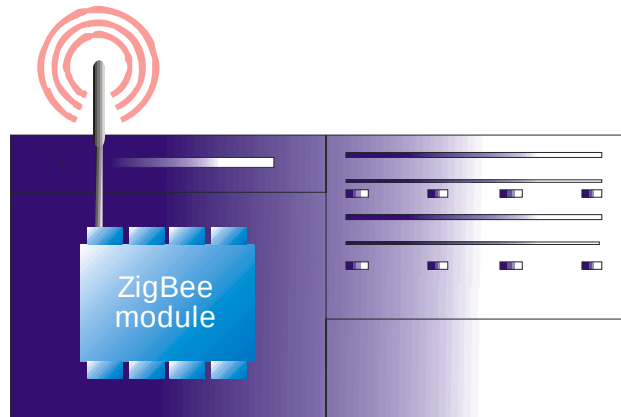
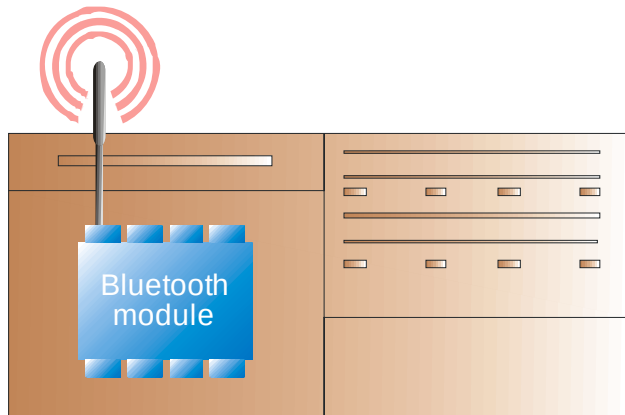
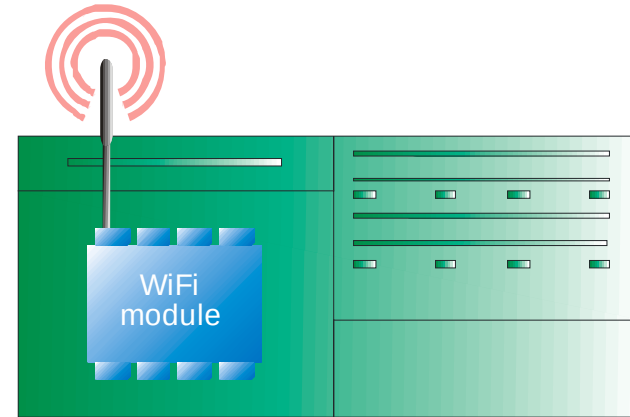
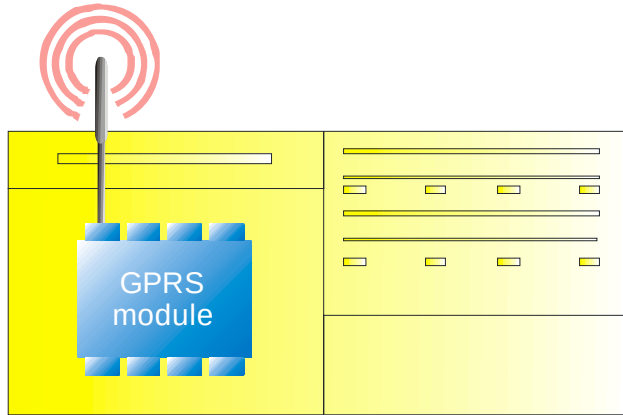
for an OPG

Agent parts and communication

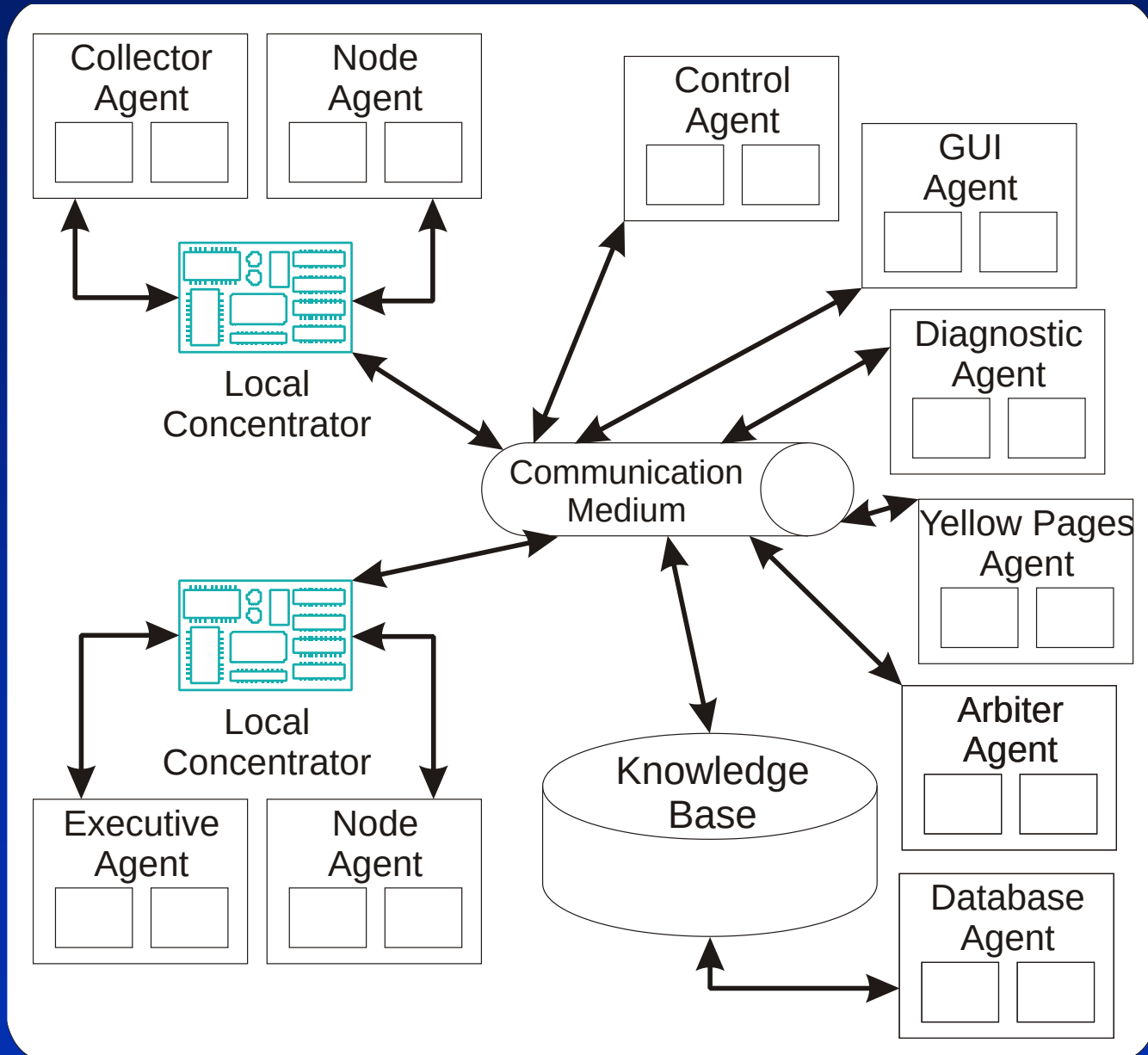


- **Kernel**
 - differs from agent to agent, depending on the specific duty of the agent: monitoring, diagnosis, control, conflict arbitration
- **Interface**
 - manages the FIPA-compliant communication with other agents or with a central knowledge base
 - generates an XML document that contains data received from the Kernel
 - sends and receives data towards/from the Interface of other agents or the knowledge base
 - decodes the received XML documents and sends the data to the Kernel.

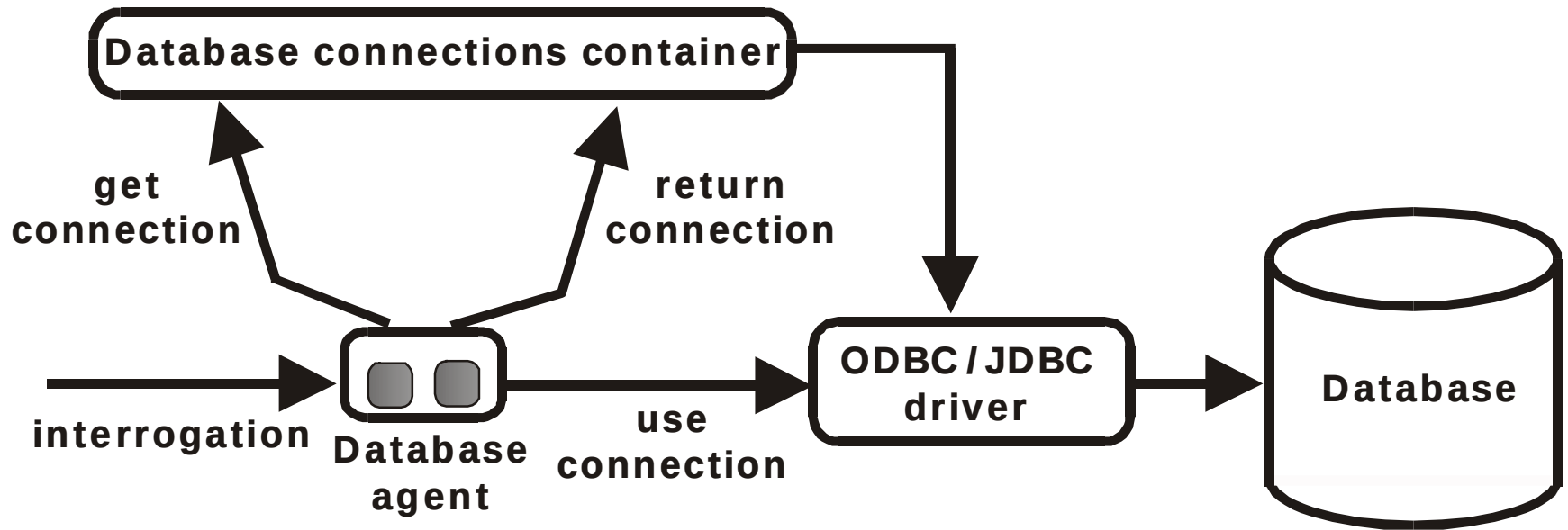
Wireless communication between the subsystems



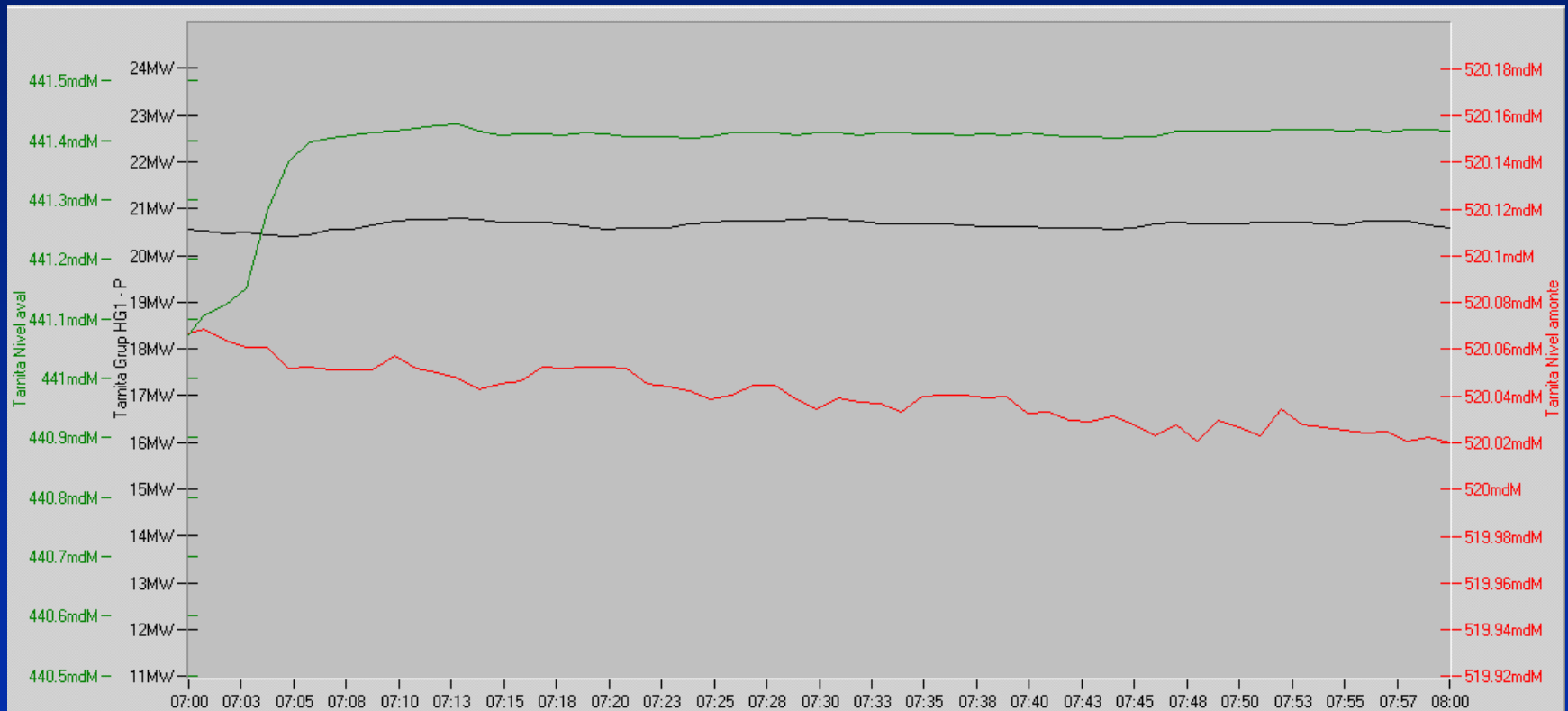
Agent roles



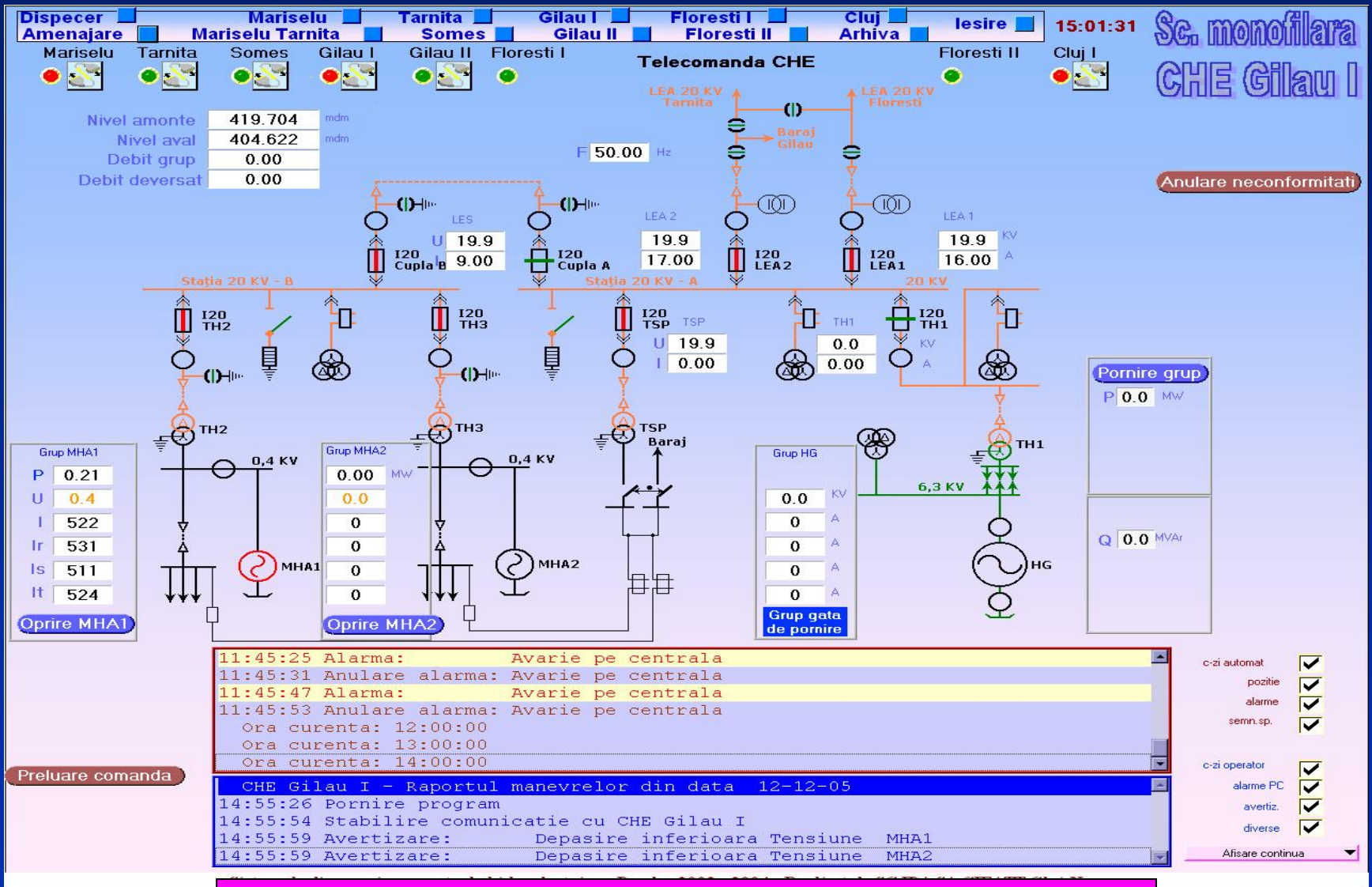
Database connection pool



Parameters of a hydropower plant generator



Hydropower plant diagram



Hydropower plant analogic and digital I/O configuration

Configurare CHE ☐ **Alocare intrari/iesiri digitale** ☐ **Alocare elem. comutare** ☐ **Dispecer** ☐
Config. contoare ☐ **Alocare intrari analogice** ☐ **Parole / Transfer fisiere** ☐ **Contoare** ☐
Config. aplicatie ☐ **Alocare achizitie / transfer** ☐ **Verificare comunicatie** ☐ **Iesire** ☐

15:50:03

Service

Alocare intrari numerice

Mariselu ☐ Tarnita ☐ Somes ☐ Gilau I ☒

Configuratii

Standard

Tiparire

Grupuri de intrari numerice

13 -24

Modificare configuratie

Mariselu

Nr semnal	Intrare Aut	Sir de cleme	Numele sub care apare semnalul	Grupa	Logica	Funct	Masura	T_el
13	I2.4 - 00	1L55	Lipsa tensiune comanda TSP5	Gen-Alarme/Av	poz	Da	0	0
14	I2.5 - 00	1L59	Lipsa tensiune operativa in statia de 20 KV	Gen-Alarme/Av	poz	Da	0	0
15	I2.6 - 00	1L64	Defect instalatie de c.c.	Gen-Alarme/Av	poz	Da	0	0
16	I2.7 - 00	1L68	Lipsa tensiune 0.4 KV servicii interne	Gen-Alarme/Av	poz	Da	0	0
17	I3.0 - 00	1L73	Gaze TSP-3	Gen-Alarme/Av	poz	Da	0	0
18	I3.1 - 00	1L77	Gaze TSP-4	Gen-Alarme/Av	poz	Da	0	0
19	I3.2 - 00	1L82	Gaze TSP-1	Gen-Alarme/Av	poz	Da	0	0
20	I3.3 - 00	1L86	Gaze TSP-2	Gen-Alarme/Av	poz	Da	0	0
21	I3.4 - 00	1L91	Presiune aer crescuta in rezervorul IP	Gen-Alarme/Av	poz	Da	0	0
22	I3.5 - 00	1L95	Nivel periculos al apei infiltrate in centrala	Gen-Alarme/Av	poz	Da	0	0
23	I3.6 - 00	1L100	Presiune minima de refulare pt apa de racire	Gen-Alarme/Av	poz	Da	0	0
24	I3.7 - 00	1L104	AAR TSP1-2 nu are conditii de functionare	Gen-Alarme/Av	poz	Da	0	0

Gilau II ☒ Floresti I ☐ Floresti II ☐ Cluj I ☐

Configuratii

Standard

Tiparire

Grupuri de iesiri numerice

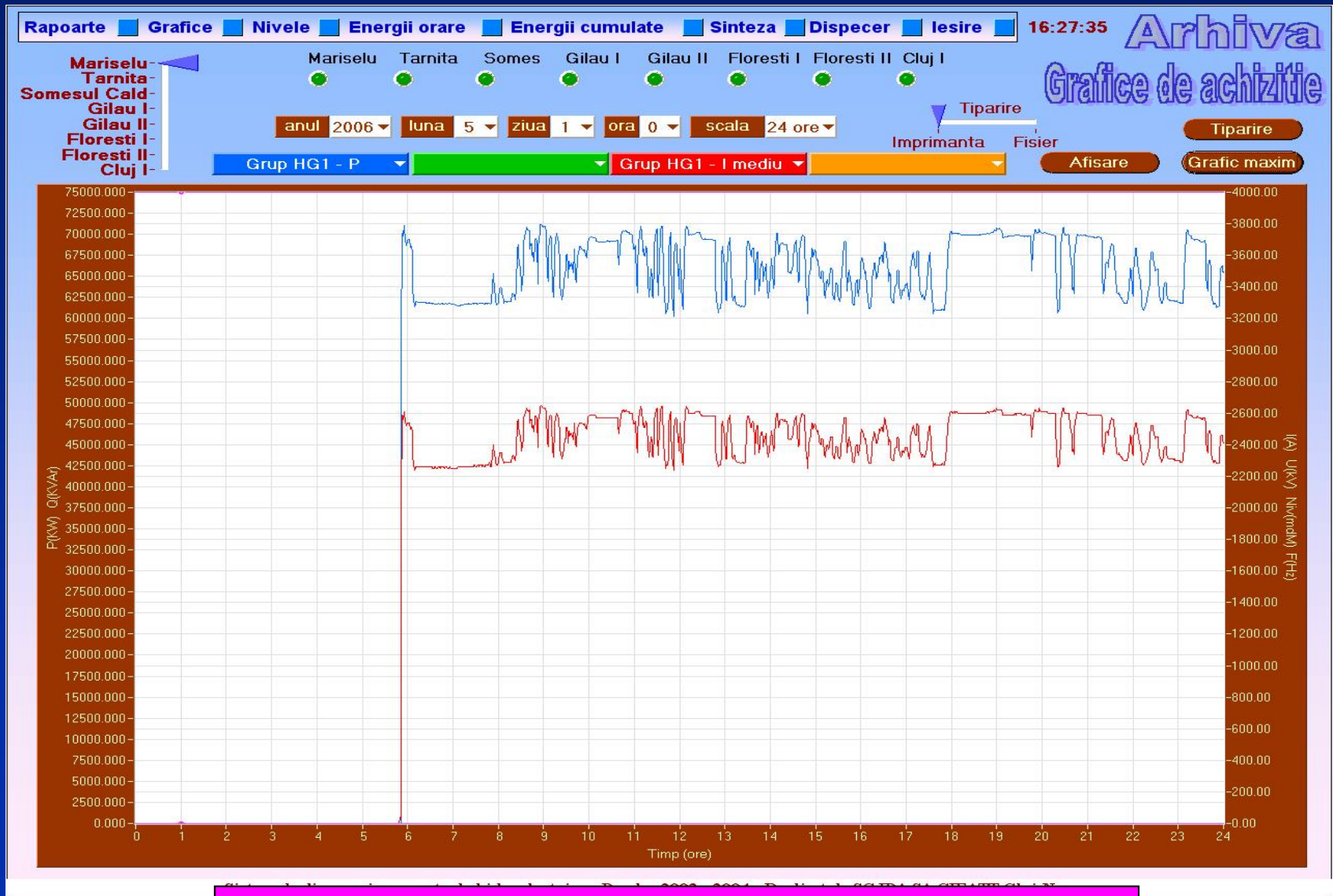
13 -24

Modificare configuratie

Nr semnal	Iesire Aut	Sir de cleme	Numele sub care apare semnalul	Grupa	Logica	Funct	Durata x10 ms	T_el / T_but
13	Q1.4	1L157	S-a dat c-da: Conectare I20 KV TSP3	Gen-Conectare	poz	Da	100	7
14	Q1.5	1L158	S-a dat c-da: Deconectare I20 KV TSP3	Gen-Deconect.	poz	Da	100	7
15	Q1.6	1L159	S-a dat c-da: Conectare I20 KV TSP4	Gen-Conectare	poz	Da	100	6
16	Q1.7	1L160	S-a dat c-da: Deconectare I20 KV TSP4	Gen-Deconect.	poz	Da	100	6
17	Q2.0	1L161	S-a dat c-da: Conectare I20 KV Fantanele 1	Gen-Conectare	poz	Da	100	8
18	Q2.1	1L162	S-a dat c-da: Deconectare I20 KV Fantanele 1	Gen-Deconect.	poz	Da	100	8
19	Q2.2	1L163	S-a dat c-da: Conectare I20 KV Fantanele 2	Gen-Conectare	poz	Da	100	9
20	Q2.3	1L164	S-a dat c-da: Deconectare I20 KV Fantanele 2	Gen-Deconect.	poz	Da	100	9
21	Q2.4	1L165			poz	Nu	0	0
22	Q2.5	1L166			poz	Nu	0	0
23	Q2.6	1L167	S-a dat c-da: Conectare I20 KV Tarnita 1	Gen-Conectare	poz	Da	100	11
24	Q2.7	1L168	S-a dat c-da: Deconectare I20 KV Tarnita 1	Gen-Deconect.	poz	Da	100	11

Observatie: ordinea semnalelor este data de programul din automatul programabil

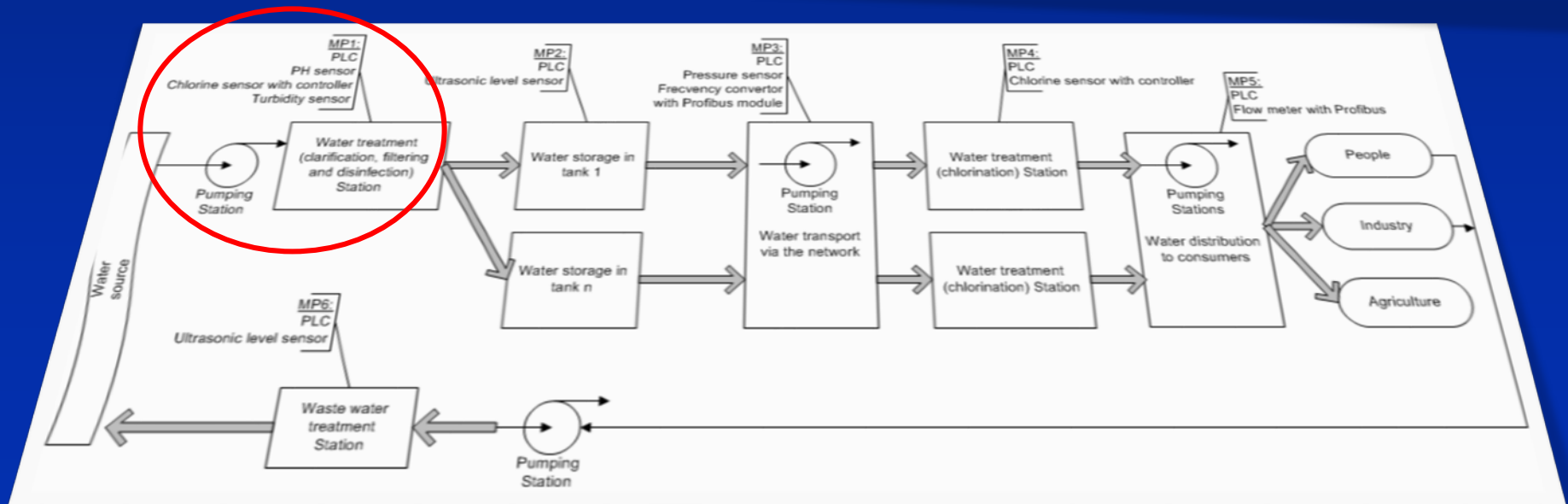
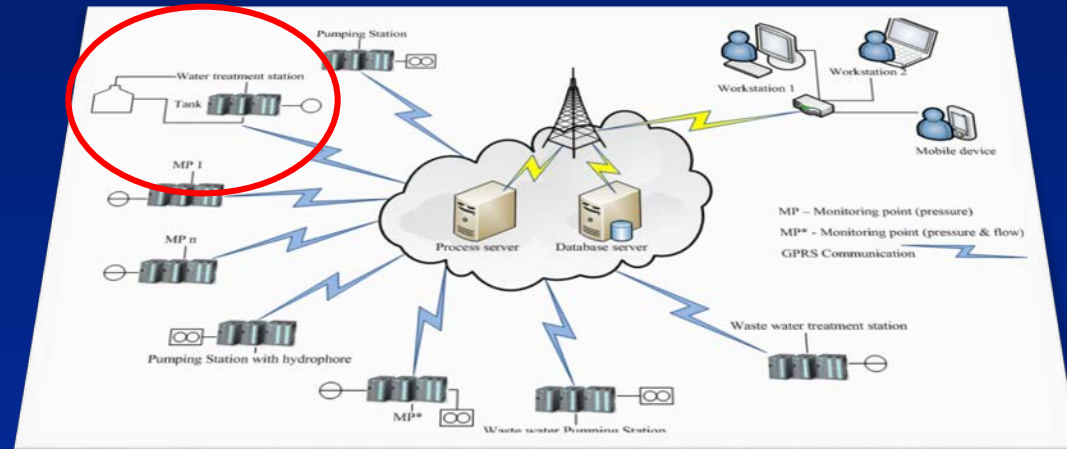
Hydropower plant report



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Cyber-Physical Water Treatment System - *is a CPS from water supply system field that incorporates monitoring and control devices to act as a distributed, dependable, and real time constrained control system*

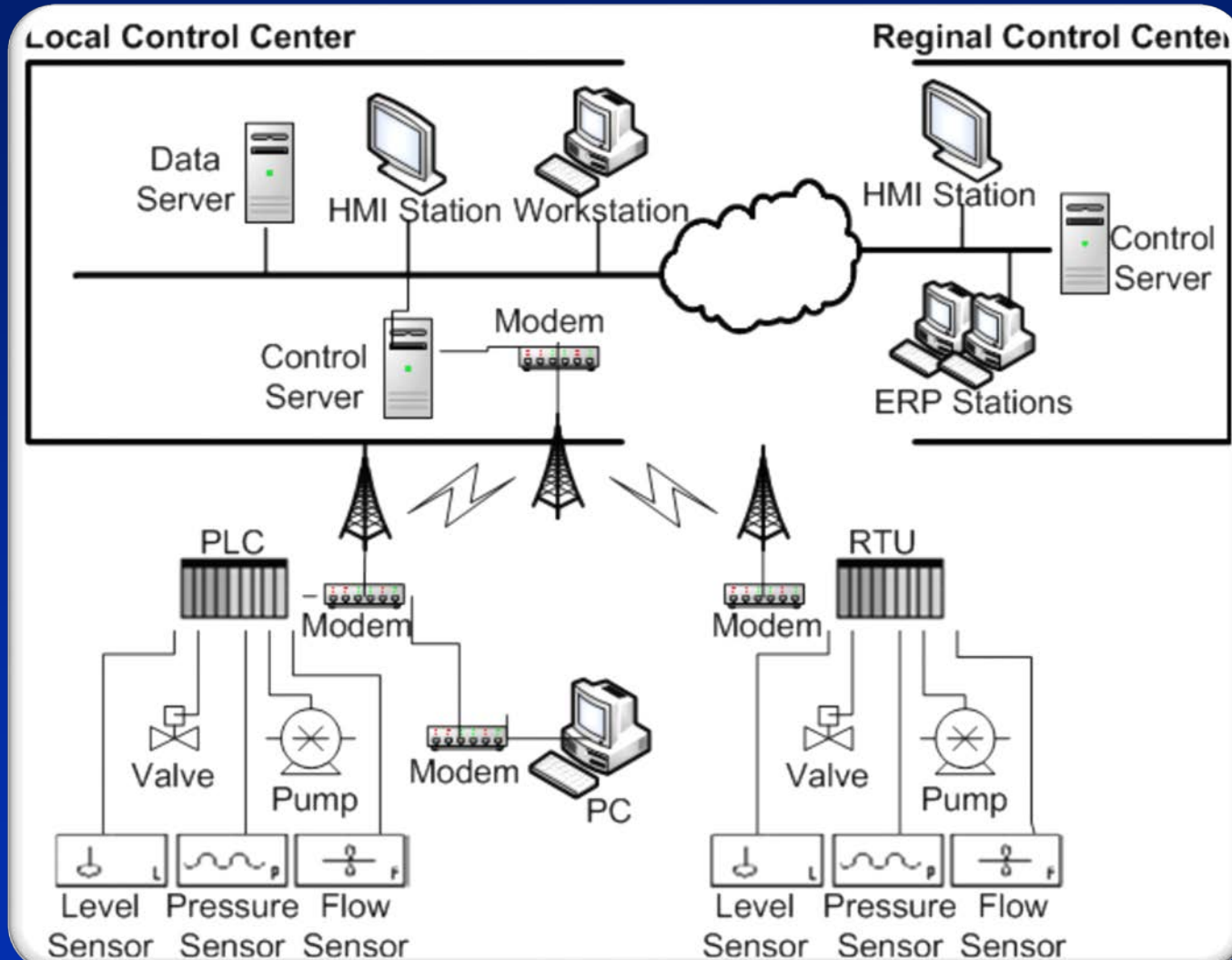




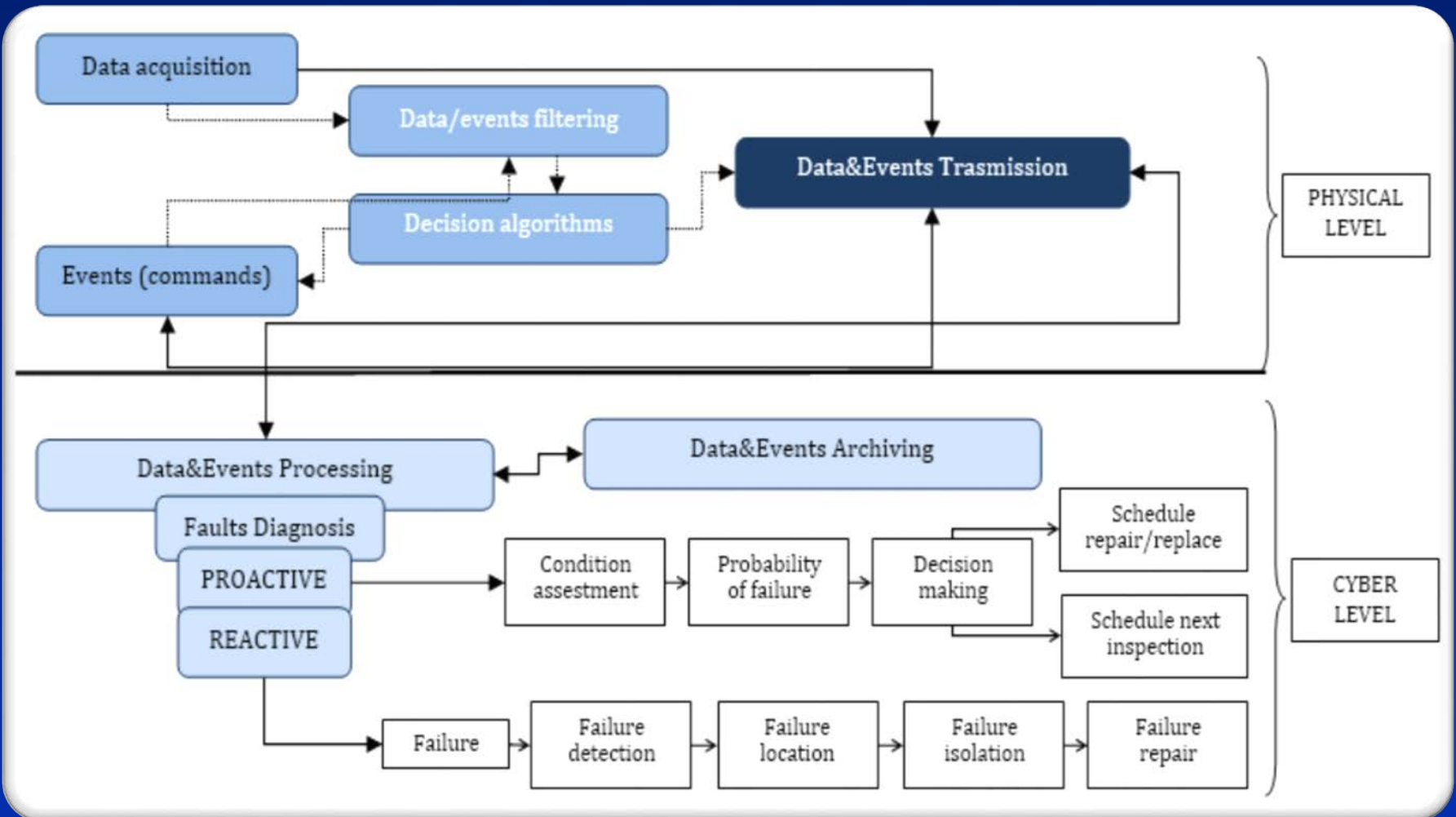
Cyber-Physical Water Treatment System characteristics

- **New generation of Information Control Systems (ICSs) for water supply domain**
- **Provides intelligent control and preventive and predictive maintenance of the physical infrastructure using the cyber components**
- **Provides support for self-healing, self-organization, self-adaptation**
- **Satisfies high reliability, robustness to unexpected conditions and high performance requirements**
- **Reduces the exploitation costs and improves the water treatment system safety in order to respond reliably under unusual conditions**
- **Rationalizes the hardware equipment and reduces the number of alarms (events)**

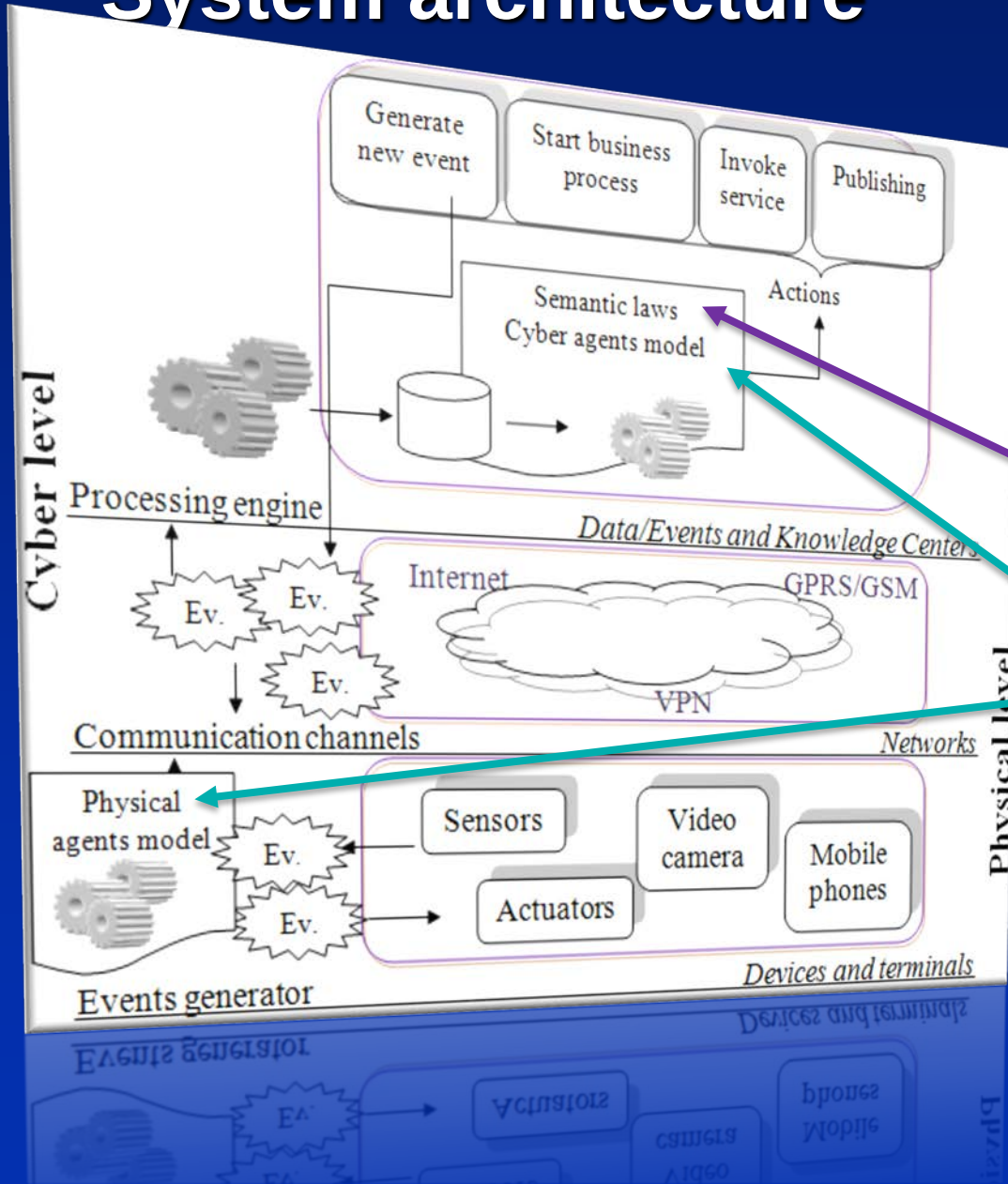
Cyber-Physical Water Treatment System structure



Cyber-Physical Water Treatment System data & events flow

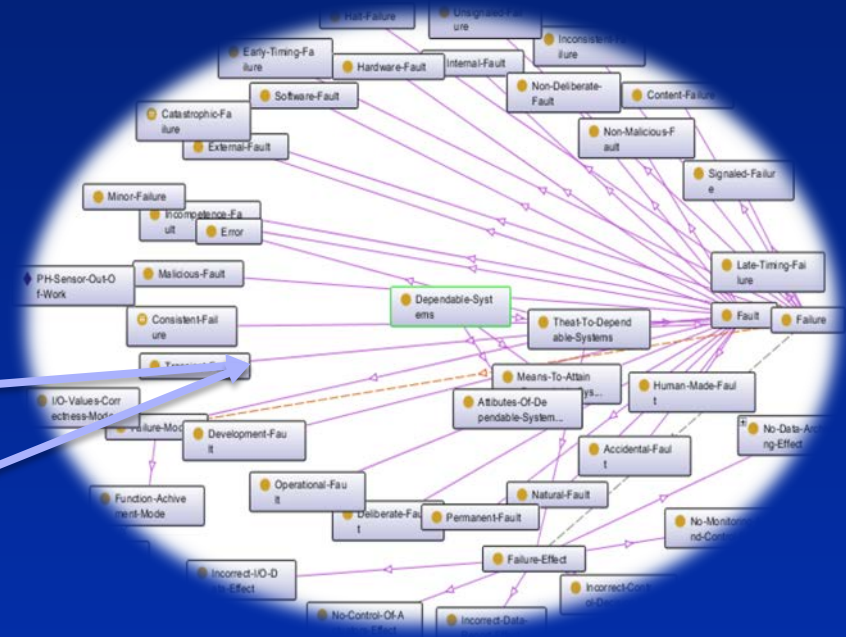
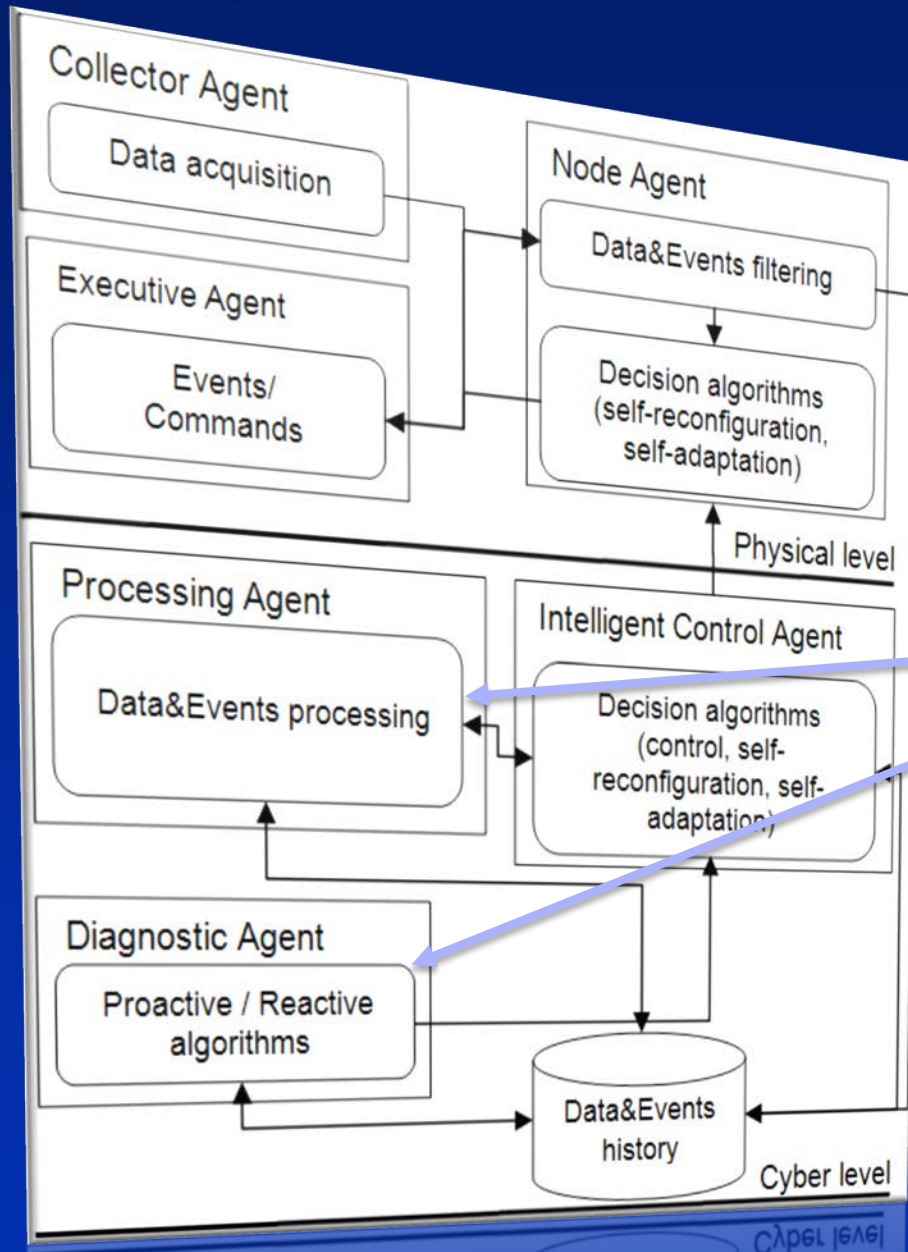


Cyber-Physical Water Treatment System architecture

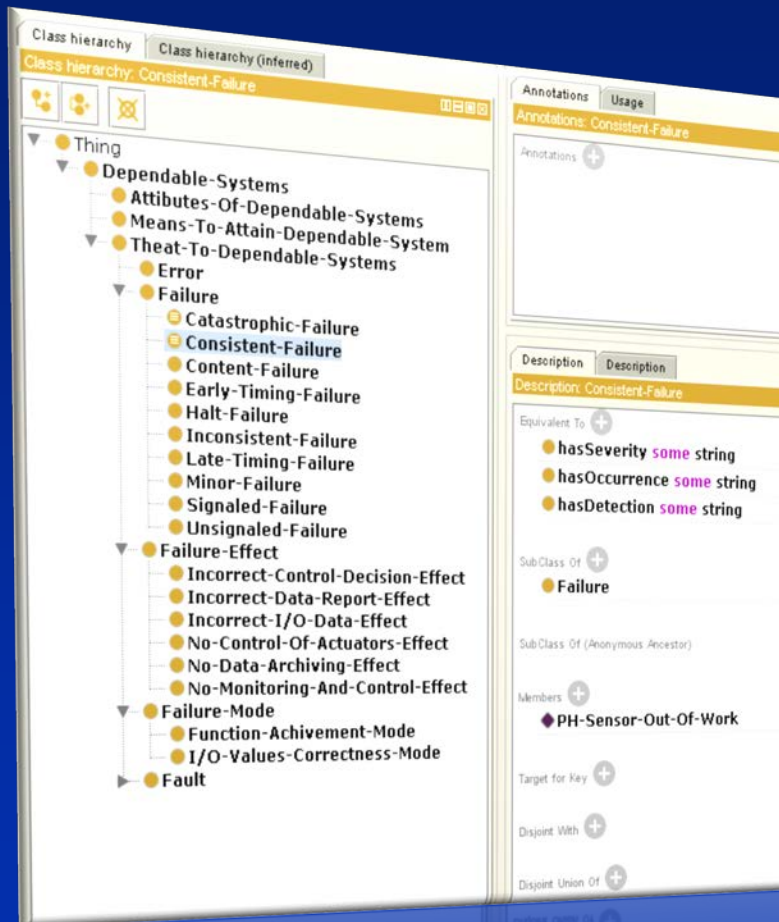


- CPS architecture based on EDA (Event-Driven Architecture)
- Dependability semantic ontology
- Multi-agent model

Agents role & Ontology

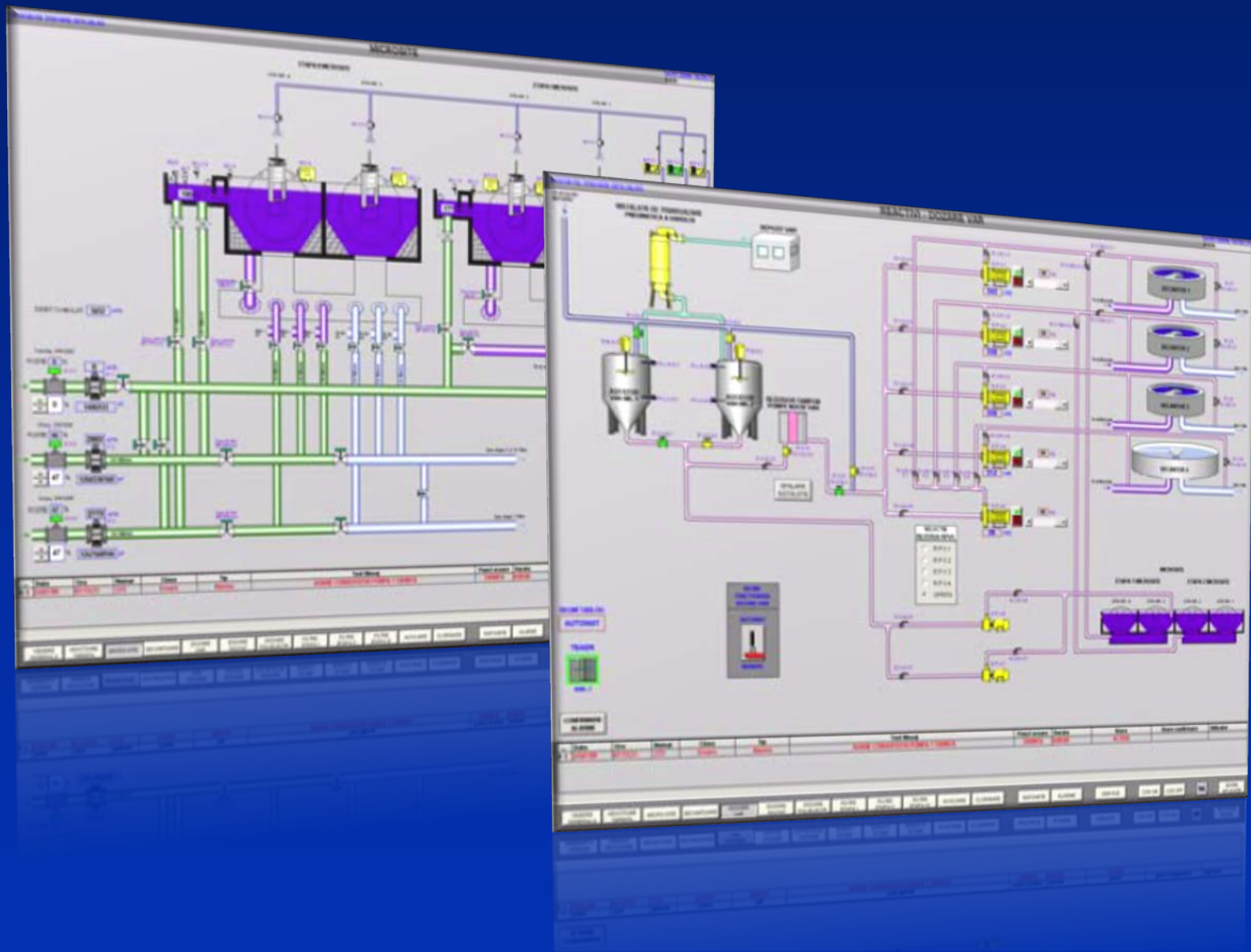


Dependability ontology



```
<owl:Restriction
  base="#Consistent-Failure">
  <owl:equivalentClass>
    <owl:Restriction>
      <owl:onProperty rdf:resource="#hasSeverity"/>
      <owl:someValuesFrom rdf:resource="#xsd:string"/>
    </owl:Restriction>
  </owl:equivalentClass>
</owl:equivalentClass>
<owl:equivalentClass>
  <owl:Restriction>
    <owl:onProperty rdf:resource="#hasOccurrence"/>
    <owl:someValuesFrom rdf:resource="#xsd:string"/>
  </owl:Restriction>
</owl:equivalentClass>
<owl:equivalentClass>
  <owl:Restriction>
    <owl:onProperty rdf:resource="#hasDetection"/>
    <owl:someValuesFrom rdf:resource="#xsd:string"/>
  </owl:Restriction>
</owl:equivalentClass>
</owl:Restriction>
```

Cyber-Physical Water Treatment System diagrams and reports



Cyber-Physical Water Treatment System diagrams and reports





Thank you

Any Questions?